

MANOJ LAMICHHANE, EIT, PE Passed

Water Resources Scientist | Hydraulic & Hydrologic Modeler | Data Science & Machine Learning
Denver, CO 80209 | (607) 663 0567 | Manoj.Lamichhane@jacks.sdstate.edu | [LinkedIn](#)

PROFESSIONAL SUMMARY

Water resources scientist with a PhD in Water Resources Engineering who works at the intersection of hydraulic modeling and data science. At Knight Piésold, performs dam breach analyses using 2D and 3D hydraulic models (FLO-2D, RiverFlow2D, FLOW-3D) and builds Python and QGIS workflows that automate model pre- and post-processing. PhD research applied machine learning and deep learning to satellite remote sensing and climate data on a USDA-funded project, using Google Earth Engine and HPC clusters for large-scale data processing. Author of 14 peer-reviewed Q1 journal papers, peer reviewer for Springer Nature and Elsevier, and an experienced communicator through 21 conference presentations and three years of university teaching.

EDUCATION

- Doctor of Philosophy in Water Resources Engineering | GPA: 4.00/4.00** **Apr 2026**
South Dakota State University, Brookings, SD
- Master of Science in Water Resources Engineering | GPA: 3.66/4.00 (WES Evaluation)** **Sep 2021**
Tribhuvan University, Nepal
- Bachelor's in Engineering | GPA: 3.77/4.00 (WES Evaluation)** **Sep 2018**
Tribhuvan University, Nepal

TECHNICAL SKILLS

Water Resources & Hydraulic Modeling

- Dam breach and flood inundation analysis
- 2D/3D hydraulic modeling: FLO-2D, RiverFlow2D, FLOW-3D, HEC-RAS
- Hydrologic modeling: HEC-HMS, SWAT, SWAT+
- Flood inundation, depth-velocity, and hazard mapping
- GIS: QGIS, ArcGIS Pro, ArcGIS Online; terrain/DEM processing
- Remote sensing: Google Earth Engine, ENVI, Pix4D, spectral indices
- Technical reporting of hydraulic modeling results

Data Science & Machine Learning

- Python (NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras); R, MATLAB, FORTRAN
- ML/DL: Random Forest, SVM, XGBoost, ANN, LSTM
- HPC: Linux clusters, parallel processing of large datasets
- Statistics: Mann-Kendall, Pettitt, OLS calibration; NSE, Willmott's d
- MLOps: reproducible end-to-end ML pipelines, automated batch training and inference on HPC (SLURM)
- Power BI, Excel (advanced modeling & VBA)

PROFESSIONAL EXPERIENCE

- Water Resources Scientist** **May 2026 – Present**
Knight Piésold
- Perform dam breach analyses to estimate breach outflow hydrographs and simulate downstream flood wave propagation, inundation extent, flow depths, velocities, and arrival times
 - Build, run, and troubleshoot 2D and 3D hydraulic models in FLO-2D, RiverFlow2D, and FLOW-3D, including model domain and grid setup, terrain and roughness inputs, boundary conditions, and breach parameters
 - Develop Python scripts to automate pre-processing of model inputs (terrain, grids, roughness, inflow hydrographs) and post-processing of model outputs into depth, velocity, and hazard rasters, summary tables, and report-ready figures
 - Use QGIS for terrain/DEM processing, model setup, and production of inundation and hazard maps for technical deliverables
 - Prepare technical documentation and communicate hydraulic modeling results to project teams
- Graduate Research Assistant** **Aug 2023 – Apr 2026**
South Dakota State University, Brookings, SD
- Developed data-driven models to estimate soil moisture, evapotranspiration, and streamflow using multi-source satellite data
 - Applied distributed and semi-distributed hydrological models to estimate water quantity, water quality, and soil moisture, achieving a 27% reduction in error for deep-layer soil moisture prediction
 - Automated data processing pipelines in Python and Google Earth Engine for real-time soil moisture prediction
 - Writing peer-reviewed journal articles and presenting findings at national and international conferences
 - Supported proposal development for NASA, USDA, and NSF and prepared technical documentation, reports, and presentations for academic and stakeholder audiences
- Water Resources Engineer** **Sep 2018 – Jun 2023**
Sentinel Engineering Consultancy, Kathmandu, Nepal

- Processed and analyzed large-scale terrain, survey, and environmental datasets across 30+ infrastructure projects, generating data products used in engineering design and permitting decisions
- Applied GIS-based terrain analysis (DEM processing, slope, flow accumulation, land use classification) to convert raw geospatial data into model-ready inputs for infrastructure planning
- Prepared technical data summaries and risk assessments for infrastructure and disaster-mitigation planning

Part-time Lecturer
Tribhuvan University, Nepal

Apr 2020 – Jun 2023

- Delivered lectures on Hydraulics, Hydrology, Fluid Mechanics, and Groundwater Engineering for undergraduate civil engineering students
- Designed and conducted lab sessions and field-based projects linking theoretical concepts with real-world water resources applications
- Mentored students on applied hydraulic and hydrologic modeling using HEC-RAS and HEC-HMS

INTERNSHIP EXPERIENCE

Summer Research Intern, High-Performance Computing Laboratory
Mississippi State University, Starkville, MS

Jun 2025 – Aug 2025

- Processed large climate, precipitation, and terrain datasets using high-performance computing (HPC) for large-scale environmental analysis
- Built an interactive ArcGIS Pro decision-support portal for evaluating environmental risk and land-surface conditions

PUBLICATIONS & PRESENTATIONS

- Authored **14 peer-reviewed research papers** in high-impact Q1 journals, focusing on hydrology, water resources engineering, machine learning, and environmental sustainability
- Delivered **21 oral and poster presentations** at national and international conferences on water resources, hydrology, and environmental engineering

LEADERSHIP & PROFESSIONAL SERVICE

- **Panelist, Student Panel, 2025 Great Plains Water Conference:** Discussed how Water Resources Research Institutes prepare graduate students for industry and academia
- **Trainer, NSF I-GUIDE 2024 Workshop, Boulder, CO:** Conducted a one-week training session on GeoAI applications to predict field-scale actual evapotranspiration
- **Peer Reviewer, Springer Nature & Elsevier (2024 – Present):** Reviewed 20+ manuscripts on water resources, hydrology, and environmental modeling

HONORS AND AWARDS

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| • Outstanding Oral Presentation , 2025 ASABE North Central Intersectional Conference, MN | 2026 |
| • 2025 ABE Graduate Scholarship Winner , South Dakota State University | 2025 |
| • Outstanding Oral Presentation , 2025 ASABE North Central Intersectional Conference, SD | 2025 |
| • Third Place, 3-Minute Thesis Competition , Department of ABE, SDSU | 2025 |
| • Best Presentation Award , AGU 2024, AGU Remote Sensing Technical Committee | 2024 |
| • Travel Award for a Conference, South Dakota State University | 2024 |
| • Outstanding Oral Presentation , ASABE North Central Regional Section Meeting, SD | 2024 |

PROFESSIONAL MEMBERSHIPS

- Member, American Geophysical Union (AGU) (2024 – Present); Committee Member, American Society of Nepalese Engineers (2024 – Present)